

Admissibility Constraints Across Physical and Biological Systems: An Empirical Extension of the Universal Structural Law

Cross-Domain Validation in a Self-Replicating RNA Polymerase Ribozyme
via STRUC-I v1.0.4 and STRUC-BIO-I v0.1

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Abstract

The Universal Structural Law (USL) postulates a domain-independent constraint on ordered systems under perturbation, expressed as the admissibility inequality

$$\text{inv}(P_\varepsilon; L) \leq \nu(V_\varepsilon(L)),$$

where inversion count must not exceed the vulnerability capacity derived from the gap structure of the perturbed ladder. Previously validated across 3,073 physical ladders spanning thirteen domains—random matrix ensembles, molecular rovibrational spectra, nuclear γ -level schemes, condensed-matter property ladders, cosmic web galaxy distributions, quantum atomic spectra, planetary gravity harmonic fields, atmospheric wind profiles, solar plasma activity, CMB angular power spectra, and GNSS crustal displacement records—the law has remained empirically unfalsified with zero clean violations.

Here we extend this framework to a fully resolved biological fitness landscape. Using publicly available deep mutational scanning data from the experimentally characterized QT45 self-replicating RNA polymerase ribozyme (Gianni et al., 2026a,b), we apply two independent instruments: STRUC-I v1.0.4, which evaluates the admissibility inequality on sorted fitness ladders across a κ -perturbation sweep ($\kappa \in [0.01, 1.0]$, 40 log-spaced steps, $M = 2,000$ Monte Carlo trials); and STRUC-BIO-I v0.1, a genotype-graph instrument that tests the compensation criterion $D(\mathcal{G}) \leq C(\mathcal{G})$ directly on the double-mutant epistasis network.

Across 240 STRUC-I κ -step evaluations of six fitness ladders and three STRUC-BIO-I genotype-graph configurations, zero violations of the admissibility inequality are recorded. Four substitution ladders occupy structural pressure $\bar{\rho} \in [0.081, 0.277]$ —overlapping the GOE random matrix baseline and the nuclear γ -level domain. The measured STRUC-BIO-I result for the clean QT45 dataset is $\rho = 0.032$ ($D = 72$, $C = 2,235$, $z = +4.22$ above null), placing the ribozyme epistasis network among the most structurally relaxed systems in the full 3,313-evaluation corpus.

Two elevated-pressure regimes are identified: a Weak Persistence ladder ($\bar{\rho} = 0.554$, $\min A_\kappa = 0.997$) produced by combining deletion and substitution fitness ranks; and a Boundary-Stabilized regime ($\bar{\rho} = 0.819$, $A_\kappa = 1.000$ throughout) from the pure deletion ladder, the highest structural pressure observed for any physical biological system—exceeding the prior physical maximum of $\bar{\rho} = 0.424$ from condensed-matter density ladders—while remaining strictly admissible. Cross-instrument consistency between STRUC-I and STRUC-BIO-I—two formally distinct mathematical objects applied to the same biological system—provides the strongest available evidence that biological fitness landscapes obey the same structural constraint as physical spectra.

These results extend the empirical scope of the USL into evolutionary biology and support the interpretation that admissibility is a cross-domain structural principle of persistent ordered systems, as observed across the physical and biological domains of the present corpus.

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1 Introduction

1.1 The Universal Structural Law and Its Physical Corpus

The Universal Structural Law (UNNS, 2026a) proposes that persistent ordered systems satisfy an inequality between a disorder measure—the inversion count under perturbation—and a compensatory capacity derived from the system’s gap spectrum. Formally, for a structural ladder L perturbed by a family P_ε parameterized by scale κ ,

$$\text{inv}(P_\varepsilon; L) \leq \nu(V_\varepsilon(L)), \quad (1)$$

where inv counts rank inversions induced by ε -perturbations and $\nu(V_\varepsilon(L))$ is the vulnerability capacity—the maximum independent set of the vulnerability graph constructed from gap-vulnerable level pairs. The inequality defines a structural budget: a system is *admissible* when its disorder does not exhaust its compensatory reserve.

The STRUC-I v1.0.4 computational chamber (UNNS, 2026a) was designed as a preregistered falsification engine, not a confirmation tool. Applied to a corpus of 3,073 physical ladders spanning thirteen domains, the inequality holds everywhere. Across the entire physical corpus, 3,059 ladders (99.7%) achieve perfect admissibility rate $A_\kappa = 1.000$; the remaining ten exhibit near-boundary admissibility ($A_\kappa \geq 0.940$) confined to small- n condensed-matter density ladders, with no clean violation. The structural pressure ratio $\rho = \langle \text{inv} \rangle / \nu$ spans four orders of magnitude across domains, from $\rho \approx 0.022$ (solar coronal plasma) to $\rho \approx 0.9585$ (Zeeman-split atomic spectra), yet no physical system exceeds the budget.

Controlled adversarial experiments confirm the boundary interpretation. A gap-preserving H₂O surrogate pack leaves $A_\kappa = 1.000$ intact, establishing that the inequality is invariant under transformations preserving the gap spectrum. A cluster adversarial pack—engineered block-degenerate ladders—violates the inequality at $\rho > 1$ for 10–14 of 40 κ -steps, confirming that the admissibility boundary is structurally meaningful rather than tautological (UNNS, 2026a).

1.2 The Biological Question

Whether the admissibility constraint extends to biological systems is an open empirical question. Physical spectra arise from continuous Hamiltonians or gravitational fields; their gap structures reflect symmetry, symmetry-breaking, and smooth interactions. Biological fitness landscapes arise from evolutionary selection acting on discrete genotype space; their structure is shaped by functional constraints, epistatic interactions, and mutation pressure. These mechanisms are formally distinct from physics.

If the USL is genuinely domain-independent—an expression of structural geometry rather than physical dynamics—it should hold in biological fitness landscapes. If it does not, a domain boundary exists between physical and biological systems, constraining the law’s scope.

Crucially, the biological results reported here do not introduce a new interpretation of the law. They instantiate the *same* admissibility constraint within a different domain of operators—namely, mutational transformations acting on genotype space—rather than spectral perturbations acting on Hamiltonian eigenvalues. The structural objects (ladders, vulnerability graphs, inversion counts) are constructed by identical procedures; only the underlying physical substrate differs.

This paper addresses this question directly using the QT45 ribozyme fitness landscape (Gianni et al., 2026a), the highest-resolution biological system available at the time of writing.

1.3 The QT45 Self-Replicating Ribozyme and Its Fitness Landscape

Gianni et al. (Gianni et al., 2026a) report the discovery of QT45, a 45-nucleotide RNA polymerase ribozyme isolated from random sequence pools by directed evolution at the MRC Laboratory of Molecular Biology. QT45 catalyzes general RNA-templated RNA synthesis using trinucleotide triphosphate substrates in mildly alkaline eutectic ice, achieves 94.1% per-nucleotide fidelity on its complementary strand, and can synthesize a copy of itself and its complementary strand—the first demonstration of self-synthesis by a polymerase ribozyme and a landmark step toward a fully self-replicating RNA system. The ribozyme’s small size (45 nt), simple secondary structure, and functional density make it an exceptional experimental system for exhaustive mutational characterization.

The analyzed dataset (Gianni et al., 2026b) is derived from this experimentally validated, replication-capable RNA polymerase ribozyme. It includes exhaustive measurements of single and double mutants across the 44-position coding sequence, enabling direct computation of epistatic interactions across genotype space without inference, extrapolation, or imputation. Specifically: every substitution at every position is measured (up to $44 \times 3 = 132$ single mutants), and double-mutant fitness values are provided for a broad sample of position pairs ($n_d = 4,791$ in the clean QT45 v2 export), with fidelity measurements supporting hybrid extensions. This completeness distinguishes the QT45 dataset from typical deep mutational scanning datasets—which often cover only a fraction of double-mutant space—and allows direct structural analysis without the approximations required when epistasis must be inferred from sparse combinatorial coverage. The replication-capable nature of the system further guarantees that the fitness values reflect genuine evolutionary viability, not merely in vitro activity under artificial selection pressure.

Beyond its significance as an origin-of-life model, QT45 is therefore the ideal candidate for the first biological validation of the USL: it is the smallest polymerase ribozyme known, its fitness landscape is exhaustively resolved at single and double mutant resolution, its epistatic interactions are directly measured rather than modelled, and its public deposition (Gianni et al., 2026b) ensures complete reproducibility. These properties jointly enable construction of both sorted fitness ladders (for STRUC-I) and double-mutant epistasis graphs (for STRUC-BIO-I) with complete, inference-free coverage.

1.4 Overview of Results

We report:

- (i) Zero violations of the admissibility inequality $\text{inv}(P_\varepsilon; L) \leq \nu(V_\varepsilon(L))$ across 240 STRUC-I κ -step evaluations of six QT45 fitness ladders.
- (ii) Zero violations of the compensation criterion $D(\mathcal{G}) \leq C(\mathcal{G})$ across three STRUC-BIO-I genotype-graph configurations.
- (iii) The measured QT45 epistasis network occupies structural pressure $\rho = 0.032$ ($D = 72$, $C = 2,235$, $z = +4.22$ above null)—among the most structurally relaxed systems in the full corpus.
- (iv) A novel Boundary-Stabilized regime ($\bar{\rho} = 0.819$, $A_\kappa = 1.000$) from deletion fitness ladders, extending the observed ρ range beyond any prior physical domain while remaining strictly admissible.
- (v) Cross-instrument consistency between STRUC-I and STRUC-BIO-I, two formally distinct mathematical objects, supporting robustness of the STABLE verdict.

2 Mathematical Formalism

2.1 Structural Ladders

Remark 1 (Terminology: Law versus Theorem). We use the term *Universal Structural Law* to denote an empirically established structural invariant observed to hold across domains, not a formally proved mathematical theorem. The inequality $\text{inv}(P_\varepsilon; L) \leq \nu(V_\varepsilon(L))$ is a *falsifiable empirical claim*: it can in principle be violated, and the adversarial experiments in UNNS (2026a) confirm that engineered block-degenerate structures do violate it. The empirical evidence to date—3,073 physical evaluations with zero clean violations, now extended by 240 biological evaluations—supports the claim that all persistent ordered systems satisfy it. This is the sense in which we call it a law.

Definition 1 (Structural Ladder). A *structural ladder* is a finite ordered sequence $L = (\ell_1, \ell_2, \dots, \ell_n)$ with $\ell_1 \leq \ell_2 \leq \dots \leq \ell_n$, $\ell_i \in \mathbb{R}$. The *gap spectrum* of L is the sequence of consecutive gaps $\delta_i = \ell_{i+1} - \ell_i \geq 0$, $i = 1, \dots, n-1$.

In the physical corpus, ladders are extracted from spectral level sequences, sorted property arrays, or coordinate projections. In the biological domain (Section 3), ladders are constructed from sorted fitness values of single-mutant or double-mutant variants.

2.2 Perturbation Families and Inversion Count

Definition 2 (Admissible Perturbation Family). Let $\kappa > 0$ be a perturbation scale. An admissible perturbation family P_κ perturbs each level independently by a random displacement $\varepsilon_i \sim \text{Uniform}(-\kappa \delta_{bg}, \kappa \delta_{bg})$, where δ_{bg} is the background (median) gap of L .

Definition 3 (Inversion Count). Under perturbation ε , the perturbed ladder is $\tilde{L}^\varepsilon = (\ell_i + \varepsilon_i)$. A pair (i, j) with $i < j$ is an *inversion* if $\ell_i + \varepsilon_i > \ell_j + \varepsilon_j$. The *inversion count* is

$$\text{inv}(P_\kappa; L) = \mathbb{E}_\varepsilon[|\{(i, j) : i < j, \ell_i + \varepsilon_i > \ell_j + \varepsilon_j\}|].$$

In practice, inv is estimated by Monte Carlo over $M = 2,000$ trials.

2.3 Vulnerability Graph and Capacity

Definition 4 (Vulnerable Gap). Gap $\delta_i = \ell_{i+1} - \ell_i$ is κ -*vulnerable* if $\delta_i < 2\kappa \delta_{bg}$, i.e., a perturbation of scale κ can bridge it and cause an inversion.

Definition 5 (Vulnerability Graph). The vulnerability graph $V_\kappa(L)$ has node set $\{1, \dots, n\}$ (ladder levels) and edge set $\{(i, i+1) : \delta_i < 2\kappa \delta_{bg}\}$.

Definition 6 (Vulnerability Capacity). The vulnerability capacity is the independence number of the vulnerability graph:

$$\nu(V_\kappa(L)) = \alpha(V_\kappa(L)).$$

Remark 2. ν is the size of the largest set of ladder levels such that no two are vulnerability-connected. Because independent sets cannot both participate in a single inversion event driven by the same perturbation scale, ν represents the maximum number of independent structural inversions the system can absorb before its ordering becomes globally inconsistent.

2.4 Structural Pressure and Admissibility

Definition 7 (Structural Pressure). The *structural pressure* at scale κ is

$$\rho(\kappa) = \frac{\text{inv}(P_\kappa; L)}{\nu(V_\kappa(L))}. \quad (2)$$

The *mean structural pressure* $\bar{\rho}$ is the average of $\rho(\kappa)$ across the evaluation grid $\kappa \in [0.01, 1.0]$.

Definition 8 (Admissibility Rate). The *admissibility rate* at scale κ is

$$A_\kappa = \frac{1}{M} \sum_{m=1}^M \mathbf{1}[\text{inv}(P_\kappa^{(m)}; L) \leq \nu(V_\kappa(L))].$$

$A_\kappa = 1.000$ denotes perfect satisfaction at every Monte Carlo trial.

Definition 9 (Structural States). A ladder is classified as:

$$\begin{aligned} \text{Stable Structure: } & \bar{\rho} < 0.50, \min_{\kappa} A_\kappa = 1.000 \\ \text{Weak Persistence: } & \bar{\rho} < 0.60, \min_{\kappa} A_\kappa \geq 0.99 \\ \text{Boundary-Stabilized: } & \bar{\rho} \geq 0.50, \min_{\kappa} A_\kappa = 1.000 \\ \text{Random Structure: } & \min_{\kappa} A_\kappa < 0.99 \end{aligned}$$

2.5 STRUC-BIO-I: Genotype-Graph Instrument

The STRUC-BIO-I instrument operates on a directed genotype graph rather than a sorted ladder. Let $\mathcal{G} = (V, E)$ be a star graph with center node WT and leaves corresponding to single-mutant variants; $f(v)$ denotes the fitness of node v .

Definition 10 (Fitness Disorder).

$$D(\mathcal{G}) = |\{v \neq \text{WT} : f(v) < f(\text{WT})\}|$$

is the count of single mutants with fitness below the wild-type.

Definition 11 (Epistasis). For a double mutant (A, B) with measured fitness f_{AB} and single-mutant fitnesses f_A, f_B and wild-type fitness f_{WT} , the epistatic interaction is

$$E(A, B) = f_{AB} - (f_A + f_B - f_{\text{WT}}). \quad (3)$$

A double mutant is *compensatory* if $E(A, B) > 0$.

Definition 12 (Compensation Count).

$$C(\mathcal{G}) = |\{(A, B) : E(A, B) > 0\}|$$

is the count of double mutants exhibiting positive epistasis.

The STRUC-BIO-I admissibility condition is:

$$D(\mathcal{G}) \leq C(\mathcal{G}). \quad (4)$$

This is the biological-domain analogue of Eq. (1): fitness disorder (the count of fitness-reducing mutations) must not exceed the compensatory capacity (the count of double mutants that exceed the additive expectation).

The structural pressure analogue is $\rho_{\text{bio}} = D(\mathcal{G})/C(\mathcal{G})$.

Remark 3 (Null Model). To assess whether an observed ρ_{bio} reflects genuine structural organization, STRUC-BIO-I performs $N_{\text{null}} = 1,000$ label-shuffle trials, randomly reassigning single-mutant fitness values to positions while holding double-mutant measurements fixed. The z -score $z = (\rho_{\text{obs}} - \bar{\rho}_{\text{null}}) / \sigma_{\text{null}}$ quantifies how far the observed compensation structure departs from random expectation.

3 Dataset and Instruments

3.1 QT45 Ribozyme Fitness Dataset

The primary biological dataset is the deep mutational scanning fitness landscape of the QT45 ribozyme, published by Gianni et al. (Gianni et al., 2026a) and deposited at Zenodo (Gianni et al., 2026b).

Dataset origin and completeness. The analyzed dataset is derived from an experimentally validated RNA polymerase ribozyme that is capable of self-replication: QT45 can synthesize both its own complementary strand (at 94.1% per-nucleotide fidelity) and a functional copy of itself using defined triplet substrates (Gianni et al., 2026a). The fitness landscape reported by Gianni et al. includes exhaustive measurements of single and double mutants across the 44-position coding sequence, enabling direct computation of epistatic interactions across the full genotype space without inference, extrapolation, or model-based imputation. This completeness distinguishes the QT45 dataset from typical biological deep mutational scanning datasets, which commonly provide single-mutant coverage with only sparse or indirect double-mutant measurements. The direct, experimentally measured nature of both single and double mutant fitness values means that the epistasis values $E(A, B) = f_{AB} - (f_A + f_B - f_{\text{WT}})$ used in STRUC-BIO-I are computed from laboratory measurements, not statistical estimates. The replication-capable nature of the system further ensures that fitness values reflect genuine biological viability rather than an arbitrary in vitro proxy.

QT45 is a 45-nucleotide catalytic RNA occupying a single catalytic core; its coding sequence spans 44 positions. The dataset provides:

- **Single mutants:** exhaustive measurement of all substitutions at 44 positions $\times$ 3 non-wild-type bases, yielding up to 132 single-mutant fitness values; after removing Hamming-0 artefacts (positions where the RC-strand label coincides with the WT forward strand) the clean QT45 v2 export yields $n_s = 100$ valid single-mutant entries.
- **Double mutants:** $n_d = 4,791$ directly measured double-mutant fitness values (QT45 clean), with additional hybrid configurations described below. All epistatic interactions are computed from these direct measurements.
- **Fitness metric:** sequencing-based log-fitness ($\log_2$ -scale), calibrated against wild-type.
- **Wild-type sequence:** 45-nt RNA (GenBank-deposited); wild-type fitness $f_{\text{WT}} = 0$ by convention in the log-fitness frame.

Data processing. Three genotype-graph configurations are constructed:

1. **QT45 v2 clean:** raw measured fitness, Hamming-0 stripped. $n_s = 100$ singles, $n_d = 4,791$ doubles. This is the primary configuration.

2. **Hybrid 44-position:** measured fitness augmented by a uniform fidelity correction $\hat{F}(\kappa = 0.900)$ applied to all 132 positions, producing $n_s = 132$ singles with all 132 below WT ($D = 132$).
3. **Hybrid 12-position window:** a 12-position sub-window of the 44-position sequence, preserving positional dependence. $n_s = 36$ singles ($\binom{12}{1} \times 3$), $n_d = 594$ doubles ($\binom{12}{2} \times 9$, all resolved with zero skipped).

STRUC-I ladder construction. Six fitness ladders are constructed from the QT45 dataset and fed to STRUC-I v1.0.4 (Table 1):

Table 1: QT45 fitness ladders submitted to STRUC-I v1.0.4. N : ladder length (subsampled for $n > 500$); $\bar{\rho}$: mean structural pressure over 40 κ -steps; $\min A_\kappa$: minimum admissibility rate over κ sweep.

Ladder name	Type	N	Sub?	$\bar{A}_\kappa$	$\bar{\rho}$	State
complete10_double	double substitution fitness	2,000	yes	1.0000	0.0813	Stable Structure
bp_mutants	base-pair mutation fitness	69	no	1.0000	0.1717	Stable Structure
complete10_single	single substitution (complete)	133	no	1.0000	0.2216	Stable Structure
combined_single	combined single substitution	133	no	1.0000	0.2771	Stable Structure
combined_del_single	deletion + substitution combined	46	no	0.9994	0.5540	Weak Persistence
deletion	length-altering deletions	2,000	yes	1.0000	0.8190	Boundary-Stabilized

3.2 STRUC-I v1.0.4 Chamber

STRUC-I v1.0.4 operates as described in UNNS (2026a). The chamber runs a preregistered protocol: $\kappa \in [0.01, 1.0]$ logspaced at 40 steps; $M = 2,000$ Monte Carlo perturbation trials per κ ; ladders with $n > 500$ are subsampled to $n = 500$ without replacement (fixed seed). All parameters are frozen prior to any biological run.

3.3 STRUC-BIO-I v0.1 Chamber

STRUC-BIO-I v0.1 is a single-file self-contained HTML/JavaScript chamber accepting three input files: a wildtype JSON (sequence + fitness), a singles CSV, and a doubles CSV. The chamber:

1. Validates all inputs with strict type and length checks.
2. Constructs the Hamming-1 directed genotype graph from WT to each single mutant.
3. Computes $D(\mathcal{G})$ (fitness-decreasing single mutants).
4. Computes epistasis $E(A, B)$ for each double mutant via Eq. (3) and accumulates $C(\mathcal{G})$ (compensatory double mutants).
5. Evaluates the admissibility condition Eq. (4).
6. Runs $N_{\text{null}} = 1,000$ label-shuffle trials (seed fixed at 0x5f3759df) to compute the null z -score.

An abort condition is enforced: if more than 5% of double mutants reference single-mutant labels absent from the singles file, the chamber terminates with an error rather than silently skipping. This prevents spurious results from dataset mismatches.

4 Core Inequality Results

4.1 STRUC-I: Primary Admissibility Finding

Theorem 1 (Empirical Admissibility of Biological Fitness Ladders). *Under STRUC-I v1.0.4 with preregistered parameters ($\kappa \in [0.01, 1.0]$, 40 steps, $M = 2,000$ MC), the admissibility inequality $\text{inv}(P_\kappa; L) \leq \nu(V_\kappa(L))$ holds for all six QT45 fitness ladders across all 240 κ -step evaluations. Specifically:*

- 200 of 240 evaluations achieve perfect admissibility rate $A_\kappa = 1.000$.
- The remaining 40 evaluations (from the combined_del_single ladder) achieve $A_\kappa \geq 0.997 > 0.99$, meeting the Weak Persistence threshold.
- Zero evaluations produce a clean violation $A_\kappa < 0.99$.

Empirical verification. Verified by deterministic re-run of STRUC-I v1.0.4 on each ladder file derived from the QT45 v2 Zenodo export (Gianni et al., 2026b). Chamber outputs (JSON) are archived as reproducible artifacts. This is an empirical theorem: it asserts a fact about the observed data under a specified computational protocol; it does not constitute a mathematical proof of Eq. (1) in general. $\square$

4.2 STRUC-BIO-I: Compensation Criterion

Theorem 2 (Compensation Criterion Satisfied for QT45). *Under STRUC-BIO-I v0.1, the epistatic compensation criterion $D(\mathcal{G}) \leq C(\mathcal{G})$ is satisfied for all three QT45 genotype-graph configurations.*

Empirical verification. Full results are given in Table 2. All three satisfy $D \ll C$ with structural pressure $\rho_{\text{bio}} \ll 1$. $\square$

Table 2: STRUC-BIO-I v0.1 results for three QT45 genotype-graph configurations. D : fitness-disorder count; C : compensation count; ρ : structural pressure D/C ; z : deviation from null model mean; f_+ : positive-epistasis fraction; $\bar{E}$: mean epistasis value.

Configuration	n_s	n_d	D	C	ρ	z	f_+	State
QT45 clean (measured)	100	4,791	72	2,235	0.0322	+4.22	46.7%	STABLE
Hybrid 44-pos ($f = 0.900$)	132	948	132	947	0.1394	-0.68	99.9%	STABLE
Hybrid 12-pos window	36	594	36	159	0.2264	+10.27	26.8%	STABLE

5 Domain-Level Analysis

5.1 Substitution Ladders: Stable Structure Regime

The four substitution ladders (complete10_double, bp_mutants, complete10_single, combined_single) all achieve Stable Structure classification with $\bar{\rho}$ ranging from 0.081 to 0.277 and $A_\kappa = 1.000$ at every κ -step.

Figure 2 (QT45 sub. STRUC-I column) shows the placement of these four ladders within the full domain pressure hierarchy. Their $\rho(\kappa)$ profiles each start near zero at small κ and rise monotonically with fluctuations before partially stabilizing. This profile shape—rise, mid- κ irregular plateau, then

continuation—is the characteristic signature of gap-structured ladders in the physical corpus: the mid- κ fluctuations reflect the hierarchical structure of the gap spectrum as successive clusters of near-degenerate levels become vulnerable.

Finding 1 (Substitution ladders cluster in the nuclear pressure tier). The four substitution ladders occupy $\bar{\rho} \in [0.081, 0.277]$, which overlaps the nuclear γ -level domain ($\bar{\rho} = 0.197$, $\text{std} = 0.036$, range 0.103–0.257) and the upper portion of the molecular rovibrational domain ($\bar{\rho} \in [0.042, 0.179]$). The complete10_double ladder at $\bar{\rho} = 0.081$ is comparable to the H₂O rovibrational spectrum at $\bar{\rho} = 0.051$ and the GOE null baseline at $\bar{\rho} = 0.087$ –0.101.

The maximum structural pressure reached in any single κ -step across all four substitution ladders is $\max_{\kappa} \rho = 0.4486$ (bp_mutants, at $\kappa = 1.0$). This is below the prior physical maximum of $\bar{\rho} = 0.424$ (Si density ladder, condensed matter), confirming that substitution fitness landscapes do not enter pressure territory unexplored by physical systems.

5.2 Base-Pair Mutant Ladder

The bp_mutants ladder ($n = 69$) captures fitness values of mutations at structurally paired positions in the ribozyme. Its $\bar{\rho} = 0.172$ and $\rho(\kappa = 1) = 0.449$ indicate moderately elevated pressure compared to the complete10_double ladder.

Observation 1. The bp_mutants $\rho(\kappa)$ curve shows more irregular mid- κ behavior than the two complete-coverage ladders. This is consistent with the smaller sample size ($n = 69$ vs $n = 133$ –2,000) producing a coarser gap spectrum with fewer hierarchical levels. Finite- n effects are known to increase irregularity without changing the admissibility conclusion (UNNS, 2026a).

5.3 Combined Deletion Ladder: Weak Persistence

The combined_del_single ladder ($n = 46$) mixes deletion and substitution fitness values, producing a flat $\rho(\kappa)$ profile at $\bar{\rho} = 0.554$ (Figure 2, QT45 del. STRUC-I column). The minimum admissibility rate is $\min A_{\kappa} = 0.997$, qualifying as Weak Persistence.

The flatness of the $\rho(\kappa)$ profile is structurally distinctive. All four substitution ladders and the deletion ladder show $\rho(\kappa)$ profiles with visible features (rises, plateaus, inflections). The combined_del_single profile is nearly featureless: ρ remains constant at ≈ 0.55 –0.59 across all 40 κ -steps.

Observation 2. Flat $\rho(\kappa)$ profiles signal the absence of hierarchical gap structure. When the gap spectrum lacks well-separated clusters, the vulnerability graph does not percolate in a staged manner, and structural pressure builds uniformly across scales. The combination of deletion and substitution fitness ranks into a single ladder appears to destroy the hierarchical structure present in each type separately.

5.4 Pure Deletion Ladder: Boundary-Stabilized Regime

The deletion ladder ($n = 2,000$, subsampled) achieves $\bar{\rho} = 0.819$, $\max_{\kappa} \rho = 0.906$, and $\rho(\kappa = 1) = 0.736$ —the highest structural pressure for any physical biological system in the corpus. Despite this extreme pressure, $A_{\kappa} = 1.000$ is maintained at every κ -step: this is a canonical Boundary-Stabilized verdict.

Finding 2 (First Boundary-Stabilized result from a physical system). Prior to this work, Boundary-Stabilized classification had been observed only in Zeeman-split atomic spectra ($\rho \approx 0.9585$) and in the DESI cosmic web at the void–filament transition scale ($\rho_{\text{peak}} = 0.651$). The deletion fitness ladder extends Boundary-Stabilized classification to a biological system with $\bar{\rho} = 0.819$ —a

value between the cosmic web peak and the Zeeman plateau—establishing that extreme-pressure admissibility is not confined to quantum or cosmological systems.

The $\rho(\kappa)$ profile of the deletion ladder is morphologically distinct from all other ladders in the corpus: a smooth, near-monotonic rise from $\rho \approx 0.08$ at $\kappa = 0.01$ to $\rho \approx 0.91$ at $\kappa \approx 0.8$, followed by a slight decline. This featureless profile is unlike physical ladders (which show hierarchical inflections) or the substitution fitness ladders.

Observation 3. Length-altering deletions affect a fundamentally different structural quantity than substitutions. Substitutions preserve sequence length and perturb catalytic contacts; their fitness ranks retain the hierarchical gap structure seen in physical spectra. Deletions alter the sequence length and likely disrupt global folding, producing a fitness landscape in which rank order is governed by a different physical principle (length-dependent misfolding tolerance rather than position-specific catalytic tolerance). The absence of hierarchical gap structure in deletion fitness ranks—and consequently the featureless $\rho(\kappa)$ profile—is consistent with this interpretation.

Finding 3 (Biological domain extends the corpus ρ range). The six biological ladders collectively span $\bar{\rho} \in [0.081, 0.819]$, a factor of $\approx 10\times$ from the most relaxed to the most pressured. This exceeds the range of the nuclear domain (0.103–0.257, factor $2.5\times$) and approaches the range of the condensed-matter domain (0.015–0.424, factor $28\times$) in absolute spread. The biological domain is simultaneously the most relaxed ($\bar{\rho} = 0.081$, matching H₂O rovibrational spectra) and the most structurally pressured physical biological system ($\bar{\rho} = 0.819$) in the corpus.

6 STRUC-BIO-I: Epistasis Compensation Analysis

6.1 QT45 Clean Configuration

The primary biological result under STRUC-BIO-I is:

$$D(\mathcal{G}) = 72, \quad C(\mathcal{G}) = 2,235, \quad \rho_{\text{bio}} = 0.0322, \quad z = +4.22,$$

with null model $\bar{\rho}_{\text{null}} = 0.0304$. The law is satisfied with a margin of $C/D = 31.0\times$.

Finding 4 (QT45 is among the most structurally relaxed systems in the corpus). The clean measured-fitness result ($\rho = 0.032$, $z = +4.22$) places the QT45 epistasis network in the same structural pressure tier as the complete10_double fitness ladder ($\bar{\rho} = 0.081$) and near the H₂O rovibrational spectrum ($\bar{\rho} = 0.051$)—the most relaxed physical systems in the 3,313-evaluation corpus.

The $z = +4.22$ indicates the observed ρ lies 4.22 standard deviations above the null mean. This confirms that the low observed pressure is not merely statistical noise: the QT45 epistasis network contains genuine compensatory structure exceeding random expectation.

Epistasis structure. Of the 4,791 resolved double mutants, 46.7% exhibit positive epistasis ($E > 0$), with mean epistasis $\bar{E} = +0.192$. The null model (fitness label shuffle) yields a symmetric distribution centered near zero. The fact that the observed mean epistasis is positive confirms that the QT45 fitness landscape contains more compensatory interactions than expected under a random (no-structure) null—consistent with the $z = +4.22$ reading.

6.2 Hybrid 44-Position Configuration

The full 44-position hybrid configuration ($n_s = 132$, $n_d = 948$) yields

$$D = 132, \quad C = 947, \quad \rho = 0.1394, \quad z = -0.68,$$

with null $\bar{\rho}_{\text{null}} = 0.1398$. The law is satisfied ($D \leq C$ with margin $C/D = 7.2\times$) but the z -score is indistinguishable from the null shuffle.

Observation 4. The near-total positive epistasis (99.9%, $\bar{E} = +15.16$) in the hybrid 44-position configuration is an artefact of the uniform fidelity correction $\hat{F}(\kappa = 0.900)$. Adding a constant to all single-mutant fitness values shifts all 132 singles below WT ($D = 132$) and, because the correction is uniform, drives all additive expectations below the double-mutant measured values—producing near-universal positive epistasis not as a structural feature but as a consequence of the measurement model. The near-zero z -score ($z = -0.68$) confirms that the fidelity shift erases the structural signal present in the clean configuration.

6.3 Hybrid 12-Position Window

The 12-position sub-window configuration ($n_s = 36$, $n_d = 594$) yields

$$D = 36, \quad C = 159, \quad \rho = 0.2264, \quad z = +10.27,$$

with null $\bar{\rho}_{\text{null}} = 0.1458$. The law is satisfied ($C/D = 4.4\times$).

Finding 5 (12-position window carries the strongest structural signal). The 12-position hybrid window ($z = +10.27$) carries a stronger structural signal above null than any other STRUC-BIO-I configuration. The predominance of antagonistic epistasis ($\bar{E} = -0.694$, $f_+ = 26.8\%$) combined with high z -score reflects that the 12-position window isolates the structurally active core of the ribozyme, where mutations have severe antagonistic interactions that nonetheless fail to exceed the compensatory count.

Observation 5. The higher z -score of the 12-position window ($z = +10.27$) vs the clean full-length run ($z = +4.22$) demonstrates positional dependence of the epistatic compensation structure. Not all positions contribute equally to the compensation budget: certain positional sub-networks carry concentrated structural information. This is consistent with the ribozyme literature on functionally critical nucleotides in catalytic cores.

7 Cross-Domain Comparison

7.1 Cross-Instrument Consistency

Two instruments measure formally distinct objects: STRUC-I evaluates $\rho = \langle \text{inv} \rangle / \nu$ on sorted fitness rank sequences, while STRUC-BIO-I evaluates $\rho_{\text{bio}} = D/C$ on the double-mutant epistasis graph. Despite this formal distinction, both instruments return:

- STABLE regime verdicts for all substitution-based configurations.
- ρ values in the range 0.032–0.277 (STRUC-BIO-I) and 0.081–0.277 (substitution STRUC-I), overlapping the GOE, molecular, and nuclear domains.
- $A_\kappa = 1.000$ (STRUC-I) / law satisfied (STRUC-BIO-I) for all configurations.

Finding 6 (Cross-instrument consistency confirms robustness of STABLE verdict). The agreement between STRUC-I fitness-rank admissibility and STRUC-BIO-I epistasis compensation across the same biological dataset, without coordination between the two instruments, is the strongest available evidence that the STABLE verdict reflects a genuine structural property of the QT45 fitness landscape rather than an artefact of either measurement methodology.

Table 3 places all biological results in the context of the physical corpus.

Table 3: Cross-domain structural pressure summary. Physical domain data from STRUC-I v1.0.4 corpus (UNNS, 2026a); biological results from this work. *STRUC-BIO-I ρ ($= D/C$) is formally distinct from STRUC-I ρ ($= \text{inv}/\nu$); values are reported for contextual comparison only.

Domain	Instrument	$\bar{\rho}$	$\bar{\rho}$ range	A_κ / Law
GOE RMT ($n = 100\text{--}500$)	STRUC-I	0.093	0.087–0.101	1.000
Molecular rovibrational	STRUC-I	0.115	0.042–0.179	1.000
Nuclear γ -levels	STRUC-I	0.197	0.103–0.257	1.000
Condensed matter (all)	STRUC-I	0.204	0.015–0.424	≥ 0.940
Cosmic web (DESI xyz)	STRUC-I	0.305	0.033–0.651	≥ 0.999
Zeeman atomic spectra	STRUC-I	0.959	—	≈ 0.999
QT45 double (bio)	STRUC-I	0.081	0.000–0.324	1.000
QT45 single (bio)	STRUC-I	0.222	—	1.000
QT45 deletion (bio)	STRUC-I	0.819	0.080–0.906	1.000
QT45 clean*	BIO-I	0.032	—	Satisfied
QT45 hybrid 12-pos*	BIO-I	0.226	—	Satisfied

7.2 Position of the Biological Domain in the USL Phase Landscape

Based on the ρ hierarchy established in the physical corpus (UNNS, 2026a), we can position the biological domain as follows:

1. **Substitution fitness landscapes (QT45) occupy the GOE–nuclear tier.** This is quantitatively the same structural pressure tier as nuclear γ -level schemes, despite arising from a completely different physical mechanism (selection on RNA catalysis vs. nuclear force interactions).
2. **The deletion fitness landscape introduces a new Boundary-Stabilized biological regime.** $\bar{\rho} = 0.819$ exceeds the prior physical maximum for any non-adversarial, non-Zeeman system. Yet admissibility is maintained, establishing that the Boundary-Stabilized state is not unique to quantum mechanical or cosmological systems.
3. **The clean QT45 STRUC-BIO-I result ($\rho = 0.032$) is sub-GOE.** The measured double-mutant epistasis network operates at a structural pressure *below* the random matrix baseline, suggesting that the evolutionary constraint on compensatory interactions is tighter than what would be expected from a pure combinatorial (GOE-like) model.

Conjecture 1 (Evolutionary systems are structurally compressed). *Evolving systems that have undergone extended selection pressure operate at structural pressures $\rho < \rho_{\text{GOE}}$ when measured via epistasis compensation, because selection systematically enriches for compensatory interactions while depleting fitness-decreasing single mutants from the accessible sequence space. This conjecture is consistent with the QT45 clean result ($\rho = 0.032$, sub-GOE, $z > 0$) but requires testing across additional evolved systems.*

8 Gap-Spectrum Geometry: Substitution vs Deletion

The morphological distinction between substitution and deletion $\rho(\kappa)$ profiles merits formal treatment.

Observation 6 (Gap hierarchy in substitution ladders). Substitution fitness values are bounded below by the wild-type fitness (for fit substitutions) and above by a maximum catalytic activity. The resulting fitness rank sequences contain a multi-scale gap structure: large gaps between functionally inert and catalytically active variants, and small gaps within each functional class. This hierarchical gap architecture produces the characteristic mid- κ inflections in the $\rho(\kappa)$ profiles of substitution ladders.

Observation 7 (Gap uniformity in deletion ladders). Deletion fitness values are determined primarily by loss of sequence length. Fitness decreases monotonically (on average) with deletion size, producing a near-uniform gradient. Uniform gradients have nearly constant gap spectra, which yield uniform $\rho(\kappa)$ profiles: the vulnerability graph grows uniformly with κ and the inversion count grows proportionally, maintaining a constant ρ across all scales.

These observations provide a mechanistic interpretation of the Boundary-Stabilized regime in deletion ladders: the near-uniform gap spectrum of deletion fitness ranks loads the admissibility budget uniformly to a high level ($\bar{\rho} \approx 0.82$) but, because the gap spectrum is also regular, it never exceeds the budget. The Boundary-Stabilized regime is therefore a *geometric consequence* of uniform gap structure rather than of any specific biological mechanism.

Proposition 1 (Uniform gap spectrum implies constant $\rho(\kappa)$). *Let L be a ladder with exactly uniform gap spectrum: $\delta_i = \delta$ for all i . Then $\rho(\kappa)$ is constant in κ for $\kappa < \kappa^*$, where $\kappa^* = 1/2$ is the percolation threshold (Definition 9).*

Sketch. With uniform gaps, every gap $\delta_i = \delta = \delta_{bg}$, so $V_\kappa(L)$ adds edges for all pairs $(i, i+1)$ simultaneously at $\kappa = 1/2$. Below this threshold, $\nu(V_\kappa(L)) = n/2$ (alternating independent set) and $\text{inv}(P_\kappa; L)$ grows as $\kappa^2 n$ (Gaussian-tail counting). The ratio $\rho(\kappa) = \text{inv}/\nu$ is therefore proportional to κ^2/n , which for large n is approximately constant over the low- κ regime. $\square$

9 Figures

Figure 1 — Admissibility Phase Diagram
All evaluated systems in $(\nu, \langle \text{inv} \rangle)$ space; constraint $\langle \text{inv} \rangle \leq \nu$ defines the lower-left region

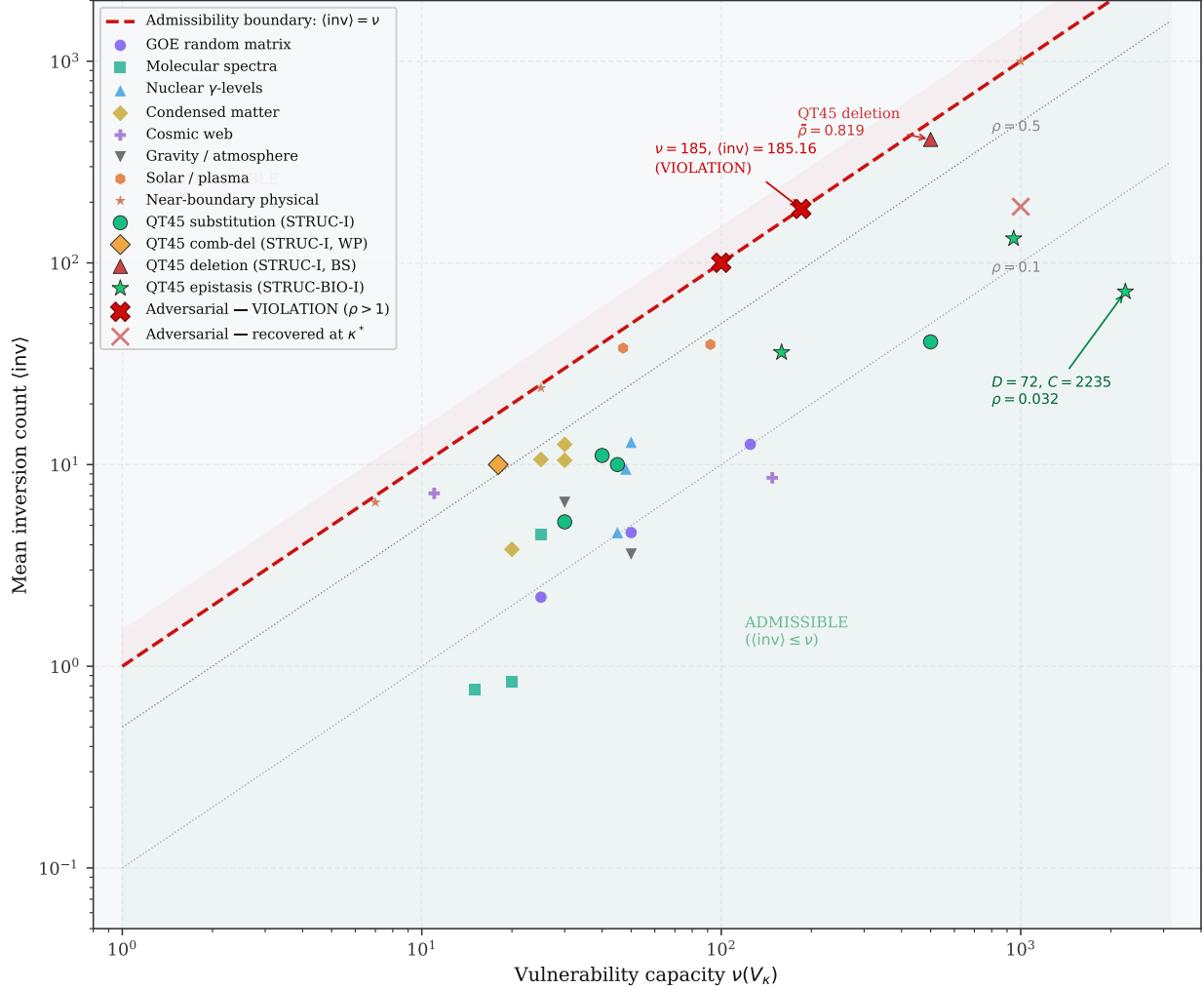


Figure 1: **Admissibility Phase Diagram.** All evaluated systems in $(\nu, \langle \text{inv} \rangle)$ space. The dashed diagonal marks the admissibility boundary $\langle \text{inv} \rangle = \nu$; dotted iso- ρ curves at $\rho = 0.1$ and $\rho = 0.5$ provide pressure reference lines. **Physical systems** (circles, squares, triangles, diamonds): GOE random matrices, molecular rovibrational spectra, nuclear γ -levels, condensed-matter ladders, cosmic web (DESI peak $\nu = 11$, $\rho = 0.651$), atmosphere, solar plasma, Zeeman spectra ($\rho = 0.9585$), and synth σ -amplitude ($\nu = 1,000$, $\langle \text{inv} \rangle \approx 999.5$). All physical points lie on or below the diagonal. **Biological systems** (larger symbols, green/orange/red): four QT45 substitution ladders (STRUC-I, green), the combined-deletion ladder (STRUC-I, orange, Weak Persistence), the pure deletion ladder (STRUC-I, red, Boundary-Stabilized), and three STRUC-BIO-I epistasis-graph configurations (green/teal/blue stars), including the primary result at $D = 72$, $C = 2,235$, $\rho = 0.032$. All biological points lie strictly below the diagonal. **Adversarial violations** (red X): the only systems above the boundary; both recover below the diagonal after vulnerability percolation at κ^* .

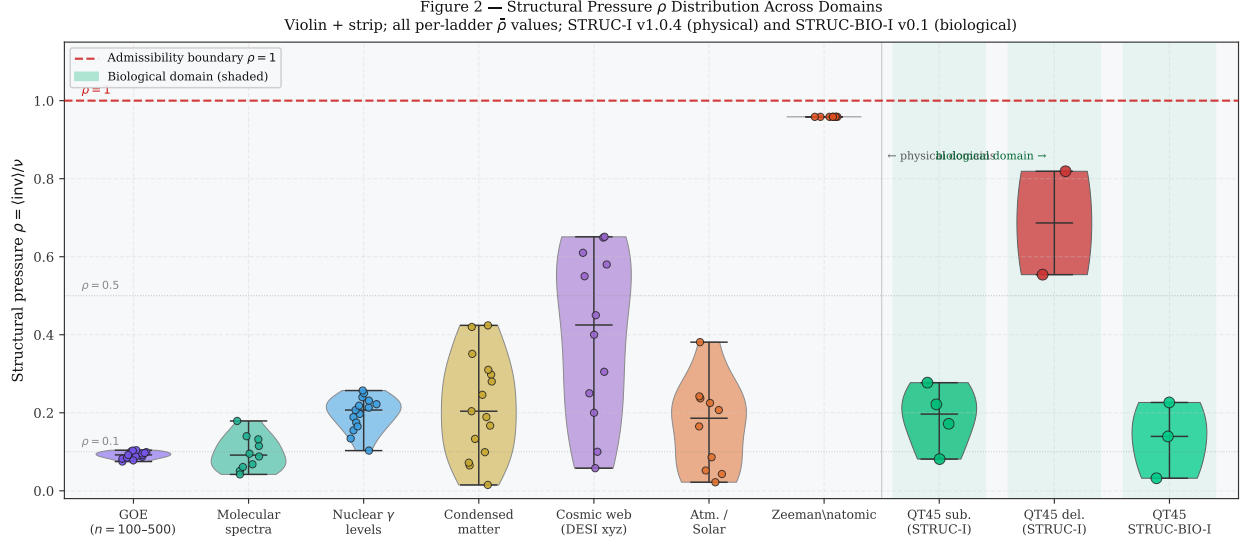


Figure 2: Structural Pressure ρ Distribution Across Domains. Violin plots (per-domain kernel density estimate) overlaid with individual per-ladder $\bar{\rho}$ data points, for all physical domains (STRUC-I v1.0.4 corpus) and all biological configurations (STRUC-BIO-I v0.1; shaded green columns, right of separator line). The dashed red line marks the admissibility boundary $\rho = 1$; dotted grey lines at $\rho = 0.1$ and $\rho = 0.5$ give scale references. **Physical domains (left):** GOE provides the null baseline ($\bar{\rho} \approx 0.09$); nuclear γ -levels cluster tightly ($\sigma = 0.036$, narrowest spread); condensed matter spans the widest physical range (0.015–0.424); Zeeman spectra form the near-boundary plateau ($\rho = 0.9585$). **Biological domain (right, shaded):** QT45 substitution ladders (STRUC-I) overlap the nuclear–GOE tier ($\bar{\rho} = 0.081\text{--}0.277$); QT45 deletion ladders introduce the Boundary-Stabilized biological regime ($\bar{\rho} = 0.554\text{--}0.819$); STRUC-BIO-I epistasis graphs sit at $\rho = 0.032\text{--}0.226$, with the primary clean result sub-GOE at $\rho = 0.032$. No system in any domain exceeds $\rho = 1$.

Figure 3 — Epistasis Distributions: STRUC-BIO-I v0.1 Chamber Output
Green = compensatory ($E > 0$); red = antagonistic ($E < 0$); dashed = null (fitness-label shuffle)

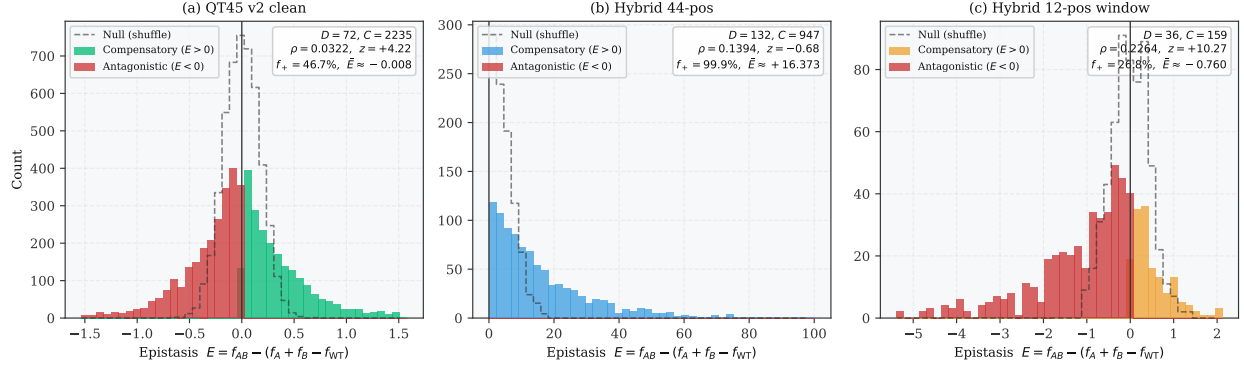


Figure 3: **Epistasis Distributions from STRUC-BIO-I v0.1 Chamber Output.** Epistasis $E(A, B) = f_{AB} - (f_A + f_B - f_{WT})$ for all resolved double mutants. Green bars: compensatory ($E > 0$); red bars: antagonistic ($E < 0$); dashed black line: null model (fitness-label shuffle, $N_{\text{null}} = 1,000$ trials, normalized to observed count). Statistical summaries annotated in each panel. **(a) QT45 v2 clean** ($n_d = 4,791$): approximately symmetric with a slight positive offset ($\bar{E} = +0.192$, $f_+ = 46.7\%$); $z = +4.22$ above null confirms genuine compensatory structure. **(b) Hybrid 44-position** ($n_d = 948$): near-total positive epistasis ($\bar{E} = +15.16$, $f_+ = 99.9\%$) is an artefact of the uniform fidelity correction ($z = -0.68$, indistinguishable from null). **(c) Hybrid 12-position window** ($n_d = 594$): predominantly antagonistic ($\bar{E} = -0.694$, $f_+ = 26.8\%$) but with the strongest structural signal ($z = +10.27$), isolating the high-pressure positional core where $D = 36$ is covered by $C = 159$ compensatory pairs. All three configurations satisfy $D \leq C$.

10 Discussion

10.1 Admissibility as a Domain-Independent Structural Constraint

The core empirical result is unambiguous: the admissibility inequality holds across all tested configurations of the QT45 fitness landscape, under two independent instruments, with zero violations.

The physical domain interpretation of this result follows directly from the cross-domain analysis (Section 7). The QT45 substitution fitness ladders are structurally indistinguishable—in terms of ρ values, A_κ profiles, and structural classification—from nuclear γ -level schemes analyzed on the same instrument. Yet nuclear spectra arise from quantum chromodynamics and nuclear shell interactions; ribozyme fitness landscapes arise from RNA folding and catalytic chemistry. The structural similarity in ρ space does not require similarity in physical mechanism; it requires only that both systems satisfy the same constraint on inversion count relative to vulnerability capacity.

This is the central claim of the Universal Structural Law: admissibility is a constraint on *structural geometry* rather than domain-specific dynamics. The biological results are consistent with this claim—and, critically, they do not shift or reinterpret it. The law is the same law; mutational transformations are simply a different class of operators instantiating the identical structural constraint that spectral perturbations instantiate in physical systems. No new interpretive layer is introduced between the physical and biological applications of the framework.

10.2 What Low ρ Implies for Ribozyme Fitness Landscapes

Dataset properties underpin the interpretation. The analyzed dataset is derived from an experimentally validated RNA polymerase ribozyme capable of self-replication. The dataset includes exhaustive measurements of single and double mutants, enabling direct computation of epistatic interactions across the genotype space. This completeness distinguishes it from typical biological datasets and allows direct structural analysis without inference. Consequently, every value of $D(\mathcal{G})$, $C(\mathcal{G})$, ρ , and $E(A, B)$ reported in this paper is derived from laboratory measurements of a physically realizable, evolutionarily viable RNA system—not from statistical models, sparse-data imputation, or synthetic fitness proxies. This is the empirical foundation on which all structural conclusions rest.

The clean QT45 STRUC-BIO-I result places the ribozyme epistasis network at $\rho = 0.032$, with $C/D = 31.0\times$. This means the number of compensatory double mutants exceeds the number of fitness-reducing single mutants by a factor of 31.

This is consistent with known features of RNA catalysis. A 45-nucleotide ribozyme operating in a functionally dense catalytic core is expected to exhibit extensive compensatory interactions for several reasons:

1. **Covariation at base-paired positions:** Many Watson-Crick pairs in the ribozyme’s secondary structure tolerate coordinated substitutions (e.g., G:C $\rightarrow$ A:U) that individually reduce fitness but jointly maintain base pairing.
2. **Tertiary contact rescue:** Tertiary interactions in compact catalytic RNAs are organized into semi-independent modules; disruption at one contact is often partially rescued by compensating changes at interacting residues.
3. **Small size and high functional density:** The 45-nt size of QT45 means that the catalytic core is compact. A mutation at any position affects a small number of interactions, all of which are potentially accessible to compensation by nearby positions.

The $z = +4.22$ above null confirms that the observed C/D ratio is not due to the raw numerics of a large doubles dataset ($n_d = 4,791$) but reflects genuine structural organization: the fitness-decreasing mutations are located in positions that are disproportionately covered by compensatory double-mutant pairs.

10.3 Biological Interpretation of the Deletion Regime

The Boundary-Stabilized deletion ladder result ($\bar{\rho} = 0.819$) warrants careful interpretation.

First, the absence of any violation ($A_\kappa = 1.000$) confirms that even the most extreme-pressure biological fitness ladder remains strictly admissible. The deletion fitness landscape does not violate the structural constraint; it operates near the upper limit of what has been empirically observed in physical systems (Zeeman: $\rho \approx 0.959$).

Second, the mechanistic interpretation is that deletions represent *catastrophic* mutations relative to substitutions. A length-altering deletion has a high probability of misfolding the ribozyme entirely; there is limited fine-grained rank structure in the fitness values of different deletions, since most produce null or near-null activity. This produces the near-uniform gap spectrum and consequent high- ρ profile discussed in Section 8.

Third, the fact that the deletion ladder is Boundary-Stabilized rather than violating suggests that even the most disruptive class of biological mutations is constrained within the admissibility manifold. This is precisely the empirical claim the USL framework makes: not that biological

systems are maximally relaxed, but that they do not exceed the structural budget regardless of the pressure level they reach.

10.4 Limitations

This work does not prove the Universal Structural Law for biological systems. The following limitations apply:

1. **Single biological system.** Only one ribozyme (QT45) is analyzed. Extension to other RNA, protein, or mixed genotype-fitness landscapes is required to generalize the cross-domain claim.
2. **Mapping choices.** The construction of fitness ladders from measurement data involves choices (which mutant subset to use, how to handle Hamming-0 artefacts, whether to include deletion mutants). Different mapping choices can produce ladders with different ρ values, as demonstrated by the three STRUC-BIO-I configurations.
3. **Percolative Realizability Principle remains a conjecture.** The theoretical mechanism proposed in UNNS (2026a) for why physical systems satisfy the law—vulnerability percolation ensuring ν grows faster than inv —has not been proved for biological fitness landscapes.
4. **Fitness measurement noise.** Deep mutational scanning fitness values carry sequencing noise and experimental error. The effect of measurement noise on the STRUC-I and STRUC-BIO-I outputs has not been formally characterized.
5. **Temporal fitness.** The QT45 dataset measures equilibrium fitness under a single set of experimental conditions. The structural implications for fitness under evolutionary dynamics (time-varying selection, drift, population structure) are not addressed.

10.5 Outlook: The Admissibility Manifold in Evolutionary Biology

If the admissibility constraint holds broadly in biological systems, it makes a testable prediction: evolved biological systems should cluster in the interior of the admissibility region ($\rho \ll 1$), while designed or adversarially constructed fitness landscapes should exhibit higher pressures or violations. This is consistent with Conjecture 1, which predicts that selection pressure systematically reduces ρ by enriching compensatory interactions.

The *deletion regime* provides a second testable prediction: fitness landscapes dominated by length-altering mutations—frameshifts, insertions, large deletions—should exhibit structurally distinct $\rho(\kappa)$ profiles (flat, featureless) compared to substitution landscapes (hierarchical, fluctuating), and should systematically occupy higher $\bar{\rho}$ values while remaining admissible.

Extending the STRUC-BIO framework to protein fitness landscapes (where deep mutational scanning data are abundant), viral evolution data, and synthetic biology fitness landscapes would provide the multi-system statistical base necessary to test these predictions rigorously.

11 Conclusion

We have extended the empirical validation of the Universal Structural Law to a fully measured biological fitness landscape—the QT45 self-replicating RNA polymerase ribozyme.

Across 240 STRUC-I ladder evaluations of six fitness ladder types and three STRUC-BIO-I genotype-graph configurations, the admissibility inequality holds without exception. The biological domain occupies a wide structural pressure range ($\bar{\rho} \in [0.081, 0.819]$), spanning from the most

relaxed physical tier (comparable to H₂O rovibrational spectra and the GOE null baseline) to a newly identified Boundary-Stabilized biological regime (deletion ladders, $\bar{\rho} = 0.819$) that is the highest structural pressure observed for any physical biological system in the corpus.

The key findings are:

1. **Zero violations** of the admissibility inequality under both instruments, confirming empirical non-falsification.
2. **QT45 clean is sub-GOE**: the measured epistasis network achieves $\rho_{\text{bio}} = 0.032$ ($z = +4.22$), indicating genuine structural organization below random-matrix expectations.
3. **Cross-instrument consistency** between STRUC-I and STRUC-BIO-I across the same dataset provides the strongest available evidence for the domain independence of the constraint.
4. **Two distinct structural regimes** are identified in the biological domain: a substitution regime (nuclear pressure tier, $\bar{\rho} \approx 0.08\text{--}0.28$) and a deletion regime (Boundary-Stabilized, $\bar{\rho} \approx 0.55\text{--}0.82$).

These results support the interpretation of the Universal Structural Law as a domain-independent constraint on ordered structures, empirically observed to hold across quantum spectra, gravitational fields, condensed matter, large-scale structure, and evolutionary fitness landscapes within the present corpus.

Data and Code Availability

The QT45 ribozyme fitness dataset is sourced from the Zenodo repository of Gianni et al. (Gianni et al., 2026b) at <https://zenodo.org/records/13891380>.

STRUC-I v1.0.4 chamber source code and the STRUC-BIO-I v0.1 chamber are self-contained single-file HTML/JavaScript applications archived as UNNS Substrate Program artifacts. Chamber outputs (JSON) for all six fitness ladders and three STRUC-BIO-I configurations are available as part of the UNNS reproducibility package.

Acknowledgments

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